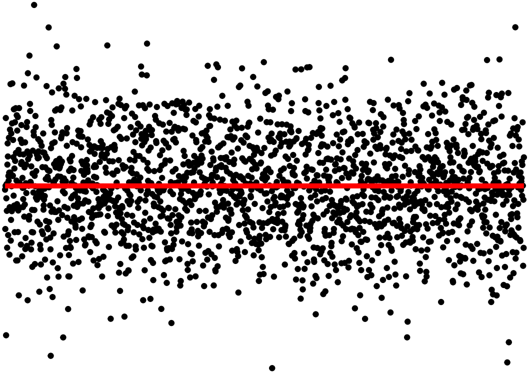
School of Mathematical Sciences | Lancaster University

# Research Overview

Rebecca Killick(rkillick@ucsc.edu)
Clemson March 2025

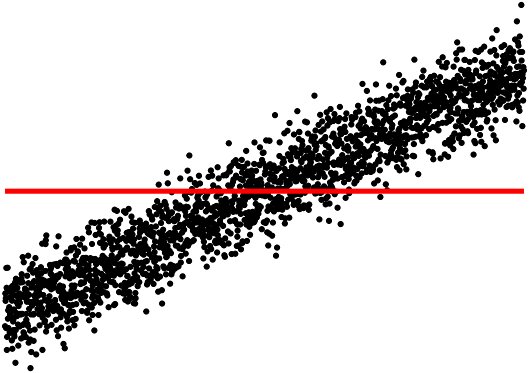
# Model development





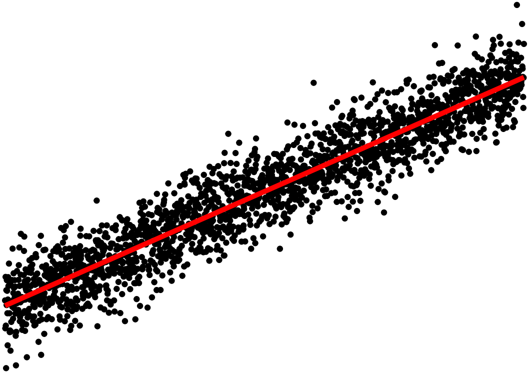
# Model development





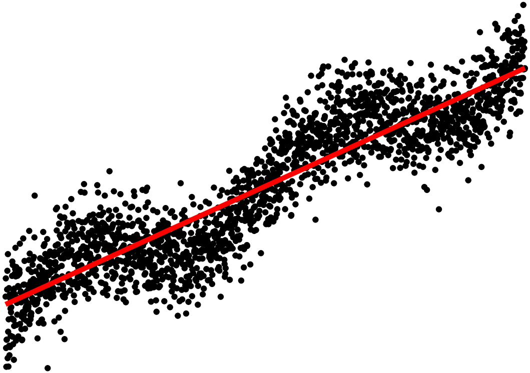
# Model development





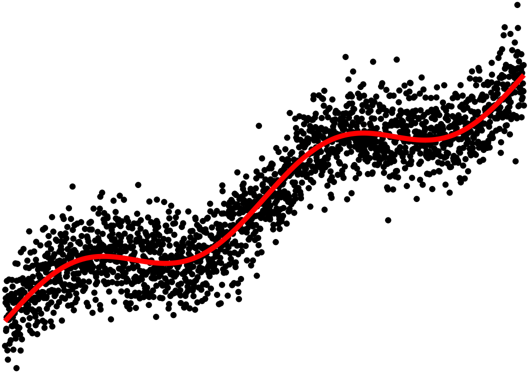
# Model development





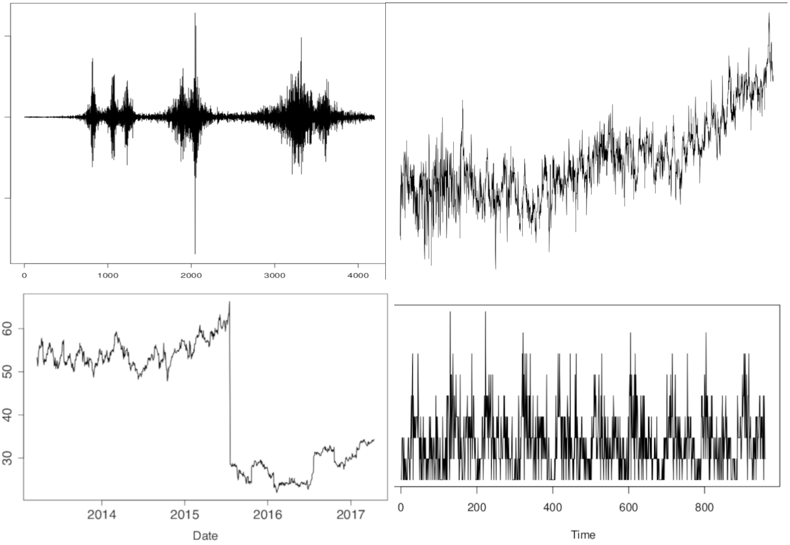
# Model development





# Data I work on





# Research Directions



Changepoints

- Search algorithms - computationally efficient multivariate
- Test statistics - changes in patterns
- Within segment model selection

Nonstationary time series - wavelets

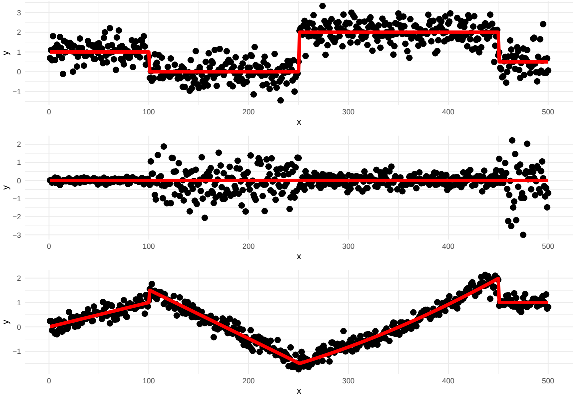
- Nonstationary mean and covariance model
- Extending stationary diagnostics to new model
- Developing classification, forecasting, methods for missingness.

# What are changepoints?



For data $y_1, \ldots, y_n$, if a changepoint exists at $\tau$, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.



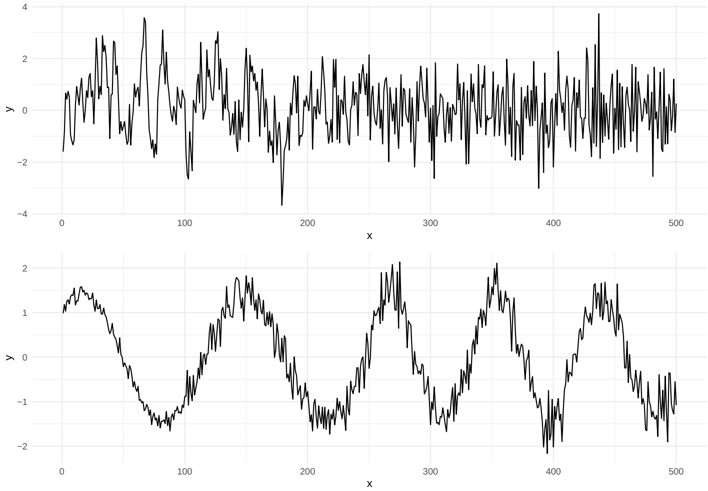
# Problem




- How many changes?
- Where are the changes? $2^{n-1}$ possible solutions!

# More complicated





# Interesting Challenges

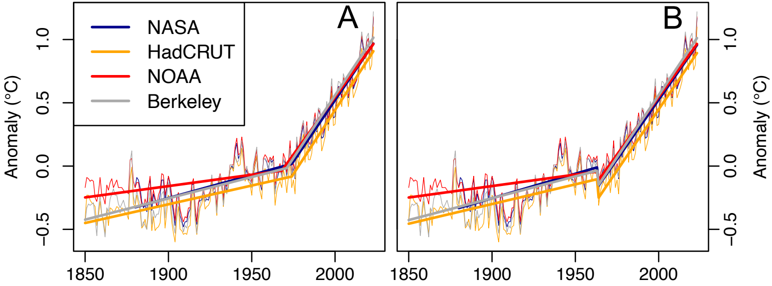


- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data

# Climate Change



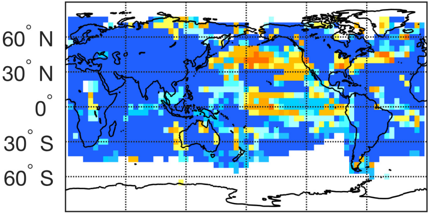
- Has there been a recent surge in global warming?
- Analysed several surface temperature time series estimates.

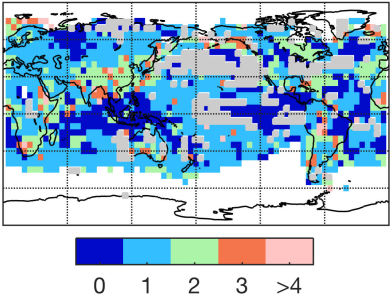



- Developed automatic model selection including trend, autocorrelation and changepoint detection
- Considered all possible models prevalent in the literature
- This indicates no surge in all series.

# Climate Dynamics



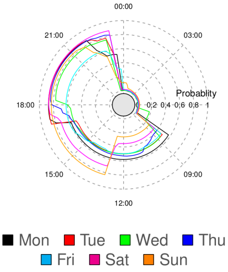


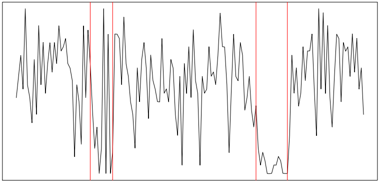



# Digital Health



- Passively monitor patients at home
- Desire to have early intervention before crisis point is reached



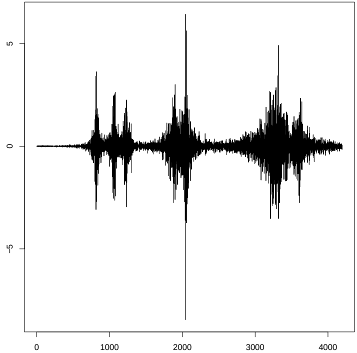



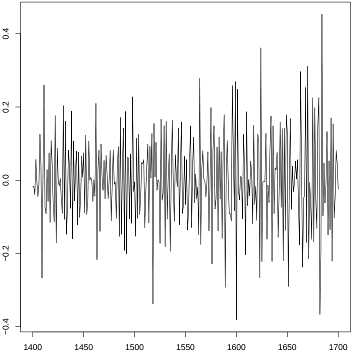
- View data as periodic, e.g., each day.
- Developed periodic changepoint detection to detect changes in activity levels within a day
- Changes in the detected routine signal early intervention

# Locally stationary



If you take a small enough region, it will appear stationary as the structure varies slowly over time.




Application areas include;

- medicine, finance
- environmental processes, e.g. wind speeds.

# T-LSW model



A nonstationary time series can be modelled as a trend locally stationary wavelet (T-LSW) process $\{X_{t,T}\}$, $t = 0, ..., T-1$, as follows:

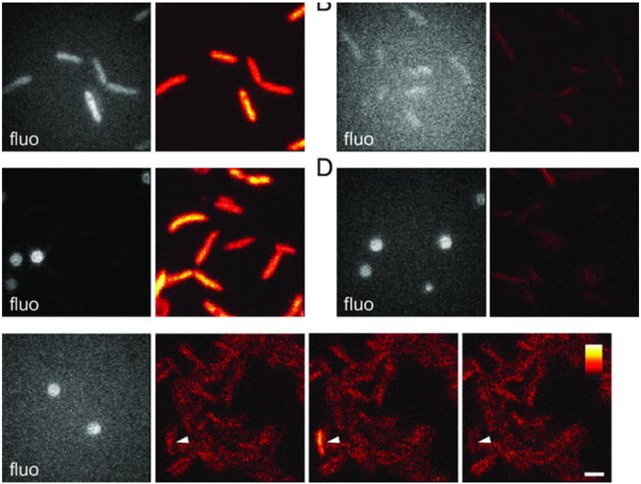
$$X_{t,T} = \mu(t/T) + \sum_{j=-\infty}^{-1} \sum_{k \in \mathbb{Z}} w_j(k/T)\psi_{j,k-t}\xi_{j,k},$$

where

- $\mu$ is a low order polynomial mean function,
- $w_j(k/T)$ are amplitudes, the smoothness of which controls the degree of nonstationarity,
- $\{\psi_{j,k-t}\}_{j,k}$ is a set of discrete non-decimated wavelets,
- $\{\xi_{j,k}\}$ is a sequence of orthonormal random variables.

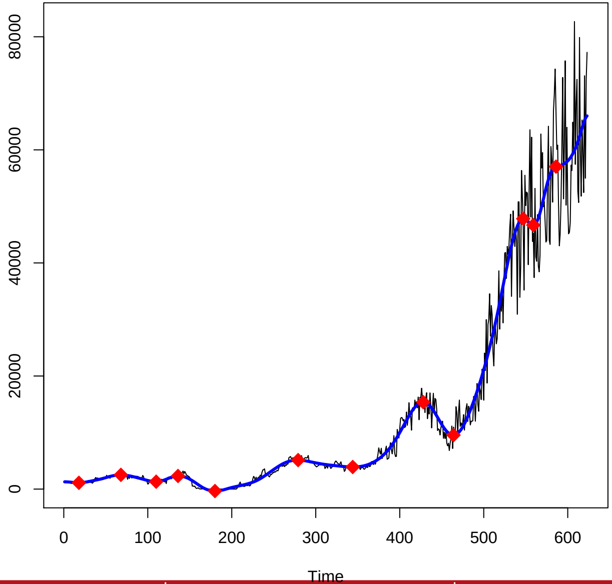
# Bioluminescence





# Bioluminescence





# Research Summary



Two main methodological research directions:

- Changepoints
- Locally Stationary time series

Three main application domains:

- Digital Health
- Environment
- Business

- I enjoy working with messy real life data …
- … as often it sparks my next research challenge